

input-queued switch—a queued packet in an input queue must wait for transfer through the fabric (even though its output port is free) because it is blocked by another packet at the head of the line. [Karol 1987] shows that due to HOL blocking, the input queue will grow to unbounded length (informally, this is equivalent to saying that significant packet loss will occur) under certain assumptions as soon as the packet arrival rate on the input links reaches only 58 percent of their capacity. A number of solutions to HOL blocking are discussed in [McKeown 1997b].

4.3.5 The Routing Control Plane

In our discussion thus far and in Figure 4.6, we’ve implicitly assumed that the routing control plane fully resides and executes in a routing processor within the router. The network-wide routing control plane is thus decentralized—with different pieces (e.g., of a routing algorithm) executing at different routers and interacting by sending control messages to each other. Indeed, today’s Internet routers and the routing algorithms we’ll study in Section 4.6 operate in exactly this manner. Additionally, router and switch vendors bundle their hardware data plane and software control plane together into closed (but inter-operable) platforms in a vertically integrated product.

Recently, a number of researchers [Caesar 2005a, Casado 2009, McKeown 2008] have begun exploring new router control plane architectures in which part of the control plane is implemented in the routers (e.g., local measurement/reporting of link state, forwarding table installation and maintenance) along with the data plane, and part of the control plane can be implemented externally to the router (e.g., in a centralized server, which could perform route calculation). A well-defined API dictates how these two parts interact and communicate with each other. These researchers argue that separating the software control plane from the hardware data plane (with a minimal router-resident control plane) can simplify routing by replacing distributed routing calculation with centralized routing calculation, and enable network innovation by allowing different customized control planes to operate over fast hardware data planes.

4.4 The Internet Protocol (IP): Forwarding and Addressing in the Internet

Our discussion of network-layer addressing and forwarding thus far has been without reference to any specific computer network. In this section, we’ll turn our attention to how addressing and forwarding are done in the Internet. We’ll see that Internet addressing and forwarding are important components of the Internet Protocol (IP). There are two versions of IP in use today. We’ll first examine the widely deployed IP protocol version 4, which is usually referred to simply as IPv4

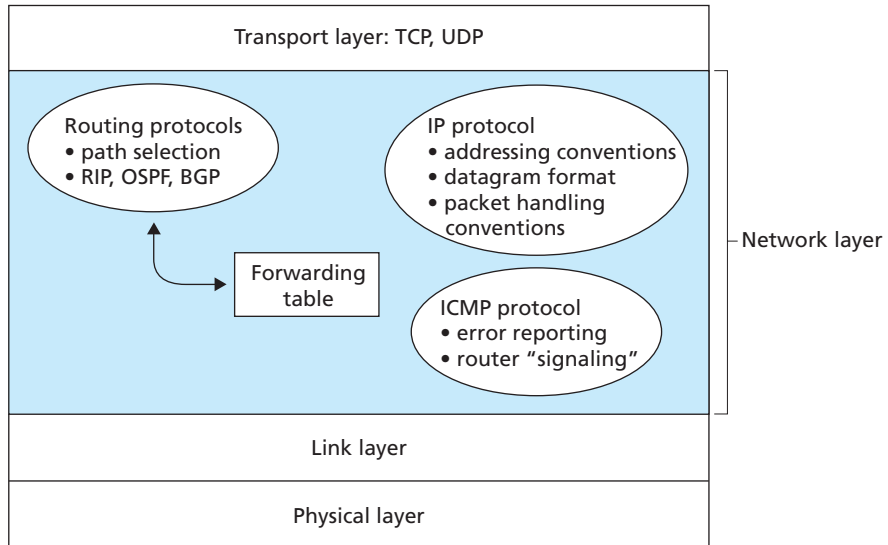


Figure 4.12 ♦ A look inside the Internet's network layer

[RFC 791]. We'll examine IP version 6 [RFC 2460; RFC 4291], which has been proposed to replace IPv4, at the end of this section.

But before beginning our foray into IP, let's take a step back and consider the components that make up the Internet's network layer. As shown in Figure 4.12, the Internet's network layer has three major components. The first component is the IP protocol, the topic of this section. The second major component is the routing component, which determines the path a datagram follows from source to destination. We mentioned earlier that routing protocols compute the forwarding tables that are used to forward packets through the network. We'll study the Internet's routing protocols in Section 4.6. The final component of the network layer is a facility to report errors in datagrams and respond to requests for certain network-layer information. We'll cover the Internet's network-layer error- and information-reporting protocol, the Internet Control Message Protocol (ICMP), in Section 4.4.3.

4.4.1 Datagram Format

Recall that a network-layer packet is referred to as a *datagram*. We begin our study of IP with an overview of the syntax and semantics of the IPv4 datagram. You might be thinking that nothing could be drier than the syntax and semantics of a packet's bits. Nevertheless, the datagram plays a central role in the Internet—every networking student and professional needs to see it, absorb it, and master it. The